

Repeated Games III: Imperfect Private Monitoring

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In the last couple of classes, we looked at repeated games with imperfect public monitoring, and studied perfect public equilibria of these games. These equilibria had a very nice recursive structure. But what happens if the outcome in each period is *not* commonly observed? In this case, we have a repeated game with imperfect *private* monitoring. Relatively little is known about the structure of equilibria in these games. These notes survey a few recent ideas on the topic.

1 Private Monitoring: An Example

To get a sense of the problems that arise when there is private monitoring, let's start by working through some variations on a simple two-period model (based on Mailath and Morris, 1998; Bhaskar and van Damme, 2002).

There are two players. In the first period, they play a prisoners' dilemma.

	C	D
C	1, 1	-1, 2
D	2, -1	0, 0

In the second period, they play a coordination game

	G	B
G	k, k	0, 0
B	0, 0	1, 1

It is assumed that $k > 2$. All the payoffs accrue after the second period, and there is no discounting. While this isn't a repeated game, it is like one in the sense that the players can try to enforce cooperation in the first period by coordinating on either a good or bad outcome later on.

1.1 Perfect Monitoring

With perfect monitoring, the following strategies will support first period cooperation:

Period 1 : Play C

Period 2 : Play G if first period outcome is (C, C) , else play B .

Clearly these strategies lead to an equilibrium in the second period, regardless of the first period outcome. In the first period, consider i 's incentives:

$$\text{Play } C \Rightarrow 1 + k$$

$$\text{Play } D \Rightarrow 2 + 1$$

Since $k > 2$, it is optimal to follow the specified strategy.

1.2 Private (Independent) Monitoring

Now suppose that first period actions (a_1, a_2) are not observed. Rather, each player i observes a signal $y_i \in \{c, d\}$ about her opponent's action. Suppose that

$$\Pr(y_i = c \mid a_j) = \begin{cases} 1 - \varepsilon & \text{if } a_j = C \\ \varepsilon & \text{if } a_j = D \end{cases}$$

If ε is small, monitoring is almost perfect. However, the striking fact is that *no pure-strategy equilibrium supports (C, C) in the first period*. Why? Observe that in the second period, i will want to play G if and only if she assigns probability $1/(k+1)$ or greater to j playing G . Consider strategies that call for each player to play C in the first period, and G in the second period if and only if $y_i = c$. If i plays C in the first period, she assigns probability $1 - \varepsilon$ to j observing c , and hence to j playing G . But then *regardless* of the signal she observes, she will want to play G , and so she won't follow the strategy.

Essentially, the problem is that i and j 's second period information is independent. So long as i cooperates in the first period, she will assign high probability to j observing a good signal regardless of her own signal. Consequently, she prefers to just keep cooperating. In short, the private monitoring means there is no way to coordinate on the punishment equilibrium in period two following a bad outcome, and on the good equilibrium otherwise. Thus, there's no way to enforce cooperation in the first period.

1.3 Private (Correlated) Monitoring

The coordination problem is lessened if the private signals are correlated rather than independent. To see this, let's consider the extreme case of perfect correlation. Suppose that $y_i = y_j = y \in \{c, d\}$, where

$$\Pr(y = c \mid (a_1, a_2)) = \begin{cases} 1 - \varepsilon & a_i = a_j = C \\ \varepsilon & \text{otherwise} \end{cases}$$

What we have now is really a game with imperfect public monitoring.

Consider strategies that call for each player to play C in the first period, and to play G in the second period if and only if they observe c . This clearly this gives an equilibrium in the second period, and checking first period incentives:

$$\begin{aligned} \text{Play } C &\Rightarrow 1 + (1 - \varepsilon)(k - 1) + 1 \\ \text{Play } D &\Rightarrow 2 + \varepsilon(k - 1) + 1 \end{aligned}$$

It is optimal to follow the specified strategy if $k \geq 1 + 1/(1 - 2\varepsilon)$, which is ensured for ε small.

More generally, if y_i and y_j are highly, but not perfectly, correlated, it will be possible to support cooperation in the first period by coordinating on different second period play depending on the signals.

1.4 Mixed Strategies

Interestingly, even if the private signals are independent, the players may be able to correlate their beliefs by playing mixed strategies in the first period. Returning to the independent signals set-up from above, consider the following strategies.

$$\begin{aligned} \text{Period 1} &: \text{ Play } C, D \text{ with probabilities } \alpha, 1 - \alpha \\ \text{Period 2} &: \text{ Play } G \text{ if and only if } a_i = C \text{ and } y_i = c. \end{aligned}$$

In the second period, i will want to play G if and only if she assigns probability $1/(k + 1)$ or greater to j playing G . She assigns this probability by conditioning on what she knows, her action a_i and her signal y_i .

$$\Pr(j \text{ will play } G \mid a_i, y_i) = \Pr(y_j = c \mid a_i) \times \Pr(a_j = C \mid y_i)$$

Applying Bayes' rule to derive the corresponding probabilities:

(a_i, y_i)	$\Pr(j \text{ will play } G)$
(C, c)	$(1 - \varepsilon) \frac{1 - \varepsilon}{\alpha(1 - \varepsilon) + (1 - \alpha)\varepsilon} \alpha$
(C, d)	$(1 - \varepsilon) \frac{\varepsilon}{\alpha\varepsilon + (1 - \alpha)(1 - \varepsilon)} \alpha$
(D, c)	$\varepsilon \frac{1 - \varepsilon}{\alpha(1 - \varepsilon) + (1 - \alpha)\varepsilon} \alpha$
(D, d)	$\varepsilon \frac{\varepsilon}{\alpha\varepsilon + (1 - \alpha)(1 - \varepsilon)} \alpha$

Aside for the (C, c) case, the probability that j will play G is of order ε . So, for sufficiently small values of ε , i will be willing to following the prescribed strategy in the second period.

Now consider i 's incentives in the first period. Her expected payoffs are:

$$\begin{aligned} \text{Play } C &\Rightarrow \alpha + (1 - \alpha)(-1) + \alpha(1 - \varepsilon)^2(k - 1) + 1 \\ \text{Play } D &\Rightarrow 2\alpha + 1 \end{aligned}$$

We can make her just indifferent between C and D by setting:

$$\alpha = \frac{1}{(k - 1)(1 - \varepsilon)^2}.$$

Note that a key to the randomization equilibrium is that player i conditions his second period behavior on the result of his first period randomization. Because of this, player j 's first period signal is informative about i 's second period behavior. This means that player j will want to condition his second period action on his first period signal, which means in turn that some incentive can be provided for i to cooperate in the first period.

2 Private Monitoring: Different Approaches

The example above illustrates some of the issues that arise with private monitoring. Let's now consider some more general approaches to the repeated game problem, and what has been shown. Most of the existing papers focus on infinite repetitions of the prisoner's dilemma (with $l \geq g > 0$):

	C	D
C	$1, 1$	$-l, 1 + g$
D	$1 + g, -l$	$0, 0$

2.1 Failure of Grim Trigger Strategies

An early paper by Compte (2002, originally 1996) shows a strong negative result. Compte argues that if private signals are independent conditional on the action profile that was played (i.e. conditionally independent), then grim trigger strategies won't work to support cooperation. The problem is that when i observes a signal that is supposed to trigger entry to the punishment phase, she will be very reluctant to initiate punishment unless she assigns high probability to j having already entered the punishment phase. But if she assigns high probability to j having already entered the punishment phase, she would not have been willing to play C to begin with. Indeed, similar to the example above, it can be shown that with conditionally independent signals, the only pure strategy equilibrium is to play D in every period. The approaches below will try to get around this in different ways.

2.2 “Almost Public” Monitoring

Mailath and Morris (2002) explore what might happen if there is private monitoring, but the signals are (highly) correlated. Their idea is to start by considering a public monitoring technology and some *strict* perfect public equilibrium under that technology (a PPE is strict if after every history, each player strictly prefers her specified action to any other). Mailath and Morris then ask whether such a perfect public equilibrium is *robust* to the introduction of a small amount of private monitoring. Let's consider their argument.

Recall that a public monitoring technology has a set of possible signals Y and a probability distribution over those signals conditional on the actions played, $\rho(y|a)$. Assume this distribution has *full support*. With private monitoring, each player i observes a signal $y_i \in Y$, and there is a probability distribution over signals $\pi(\mathbf{y}|a)$. The private monitoring technology is said to be ε -close to ρ if for all a and $y \in Y$, $|\pi(y, \dots, y|a) - \rho(y|a)| < \varepsilon$. This is MM's notion of “almost public” monitoring.

In the prisoner's dilemma, an example of a public monitoring technology has $Y = \{\underline{y}, \bar{y}\}$, and

$$\rho(\bar{y} \mid a_1, a_2) = \begin{cases} p & \text{if } (C, C) \\ r & \text{otherwise} \end{cases}.$$

Mailath and Morris consider strategies that can be represented as finite automata. In the case of the prisoners' dilemma, let's consider strategies defined by a two-state automaton with state space $W = \{w_C, w_D\}$, where i

is supposed to play C in state w_C and D in state w_D . A symmetric strategy profile is defined by an initial state, and a transition function $\sigma(yw)$. For example, grim trigger has initial state w_C , and transition $\sigma(yw) = w_C$ if and only if $w = w_C$ and $y = \bar{y}$. Mailath and Morris also consider the following example, where the initial state is w_C and

$$\sigma(yw) = \begin{cases} w_C & \text{if } y = \bar{y} \\ w_D & \text{if } y = \underline{y} \end{cases}.$$

The strategies that result are a strict equilibrium of the public monitoring game if $\delta \geq \frac{1}{3(p-r)}$. Checking this requires showing that if i, j are in state w_C , then i strictly wants to play C , while if i, j are in state w_D , i strictly prefers D .

When we introduce private monitoring, it is natural to think of each player having her own *private* state $w_i \in \{w_C, w_D\}$, where i is supposed to play C if $w_i = w_C$ and play D if $w_i = w_D$. We can consider the same strategies — same initial state and transitions functions — only now the transitions operate on the private states, so $\sigma(y_i w_i)$. To check the relevant incentives, we need to know the probability that i will assign to her opponent being in different states after each history. (Of course, with public monitoring, i and j are always in the same state). If, regardless of what has happened, i believes that her opponents are in the same state as she is with very high probability (say $1 - \zeta$), she will not want to deviate given that the original equilibrium was strict (since the private monitoring will just add a negligible term of order ζ to each side of the incentive constraint). MM try to identify conditions on strategies that will ensure that players end up almost certain of being in the same state at every point in time.

Interestingly, it turns out that one condition that ensures the necessary correlation across private states is that strategies depend only on a finite history of signals. The transition given above depends only on the last signal, so it satisfies this condition, and hence the strategies remain an equilibrium with almost-public monitoring. Grim trigger, however, depends on the whole infinite history of play. MM point out that with perfect monitoring, one can prove the folk theorem with finite history strategies. Thus, they obtain a folk theorem for games with almost-public, almost-perfect monitoring.

2.3 Initial Randomization

While Mailath and Morris try to identify which public monitoring strategies can be transported directly into games with a little bit of private monitoring, Sekiguchi (1997) explicitly constructs strategies for the repeated prisoner's

dilemma with private monitoring, and shows that they allow cooperation provided that the monitoring is almost-perfect. Sekiguchi requires both players to randomize at the beginning of the game between playing grim trigger, and playing defect in every period. It turns out this is an equilibrium for intermediate discount factors provided monitoring is sufficiently accurate.

Sekiguchi has to tackle two issues to verify the equilibrium. First, if i has been cooperating, and has always received good signals, will she still want to cooperate? It turns out she will if δ is high enough and monitoring is sufficiently accurate. Second, will i want to defect as soon as a bad signal is received? If she receives the bad signal early on (e.g. the first period), she can conclude that her opponent started with the always defect strategy, so she is happy to defect. On the other hand, if she receives the defect signal only after some time, the most likely events are that either (i) the signal is erroneous, or (ii) j received a bad signal in the previous period and has just entered the punishment phase. These events have equal probability, but Sekiguchi manages to show that if δ is not too high, i will be willing to experiment by defecting. Thus, he obtains cooperation with almost-perfect (but not necessarily almost-public) monitoring.

2.4 Belief-Free Equilibria

Piccione (2002) and Ely and Valimaki (2002) consider a very different approach that is based on ongoing randomization. These authors observe that the above approaches require keeping track of players beliefs about what the others are doing over time. This can be very complicated. Their approach is to construct equilibria for games with almost-perfect (but not necessarily correlated) signals in which the players are *indifferent* between C, D at *every point in time*.

As an example, consider the repeated prisoner's dilemma:

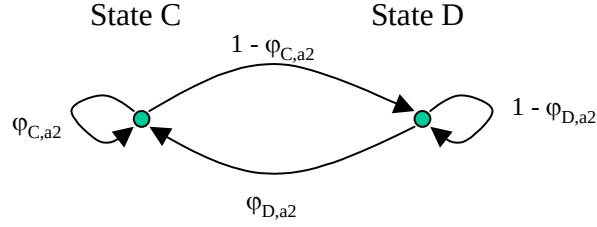
	C	D
C	1, 1	-1, 2
D	2, -1	0, 0

where, after each period, player i observes a signal $Y_i \in \{c, d\}$ and:

$$\Pr(Y_i = c) = \begin{cases} 1 - \varepsilon & \text{if } a_i = C \\ \varepsilon & \text{if } a_i = D \end{cases}$$

The P-EV approach is to construct sequential equilibria in which each player are *indifferent* between C, D at *every point in time* – i.e. after every possible history.

To see how this might be done, consider the perfect monitoring version of the repeated prisoner's dilemma ($\varepsilon = 0$). Following EV, consider strategies as follows. Player i plays C in the first period, and in any subsequent period t , plays a mixed strategy that depends only on the outcome at $t - 1$. Let φ_{a_i, a_j}^i denote the probability she plays C at t conditional on outcome (a_i, y_i) at $t - 1$. We can think of this strategy as a two-state machine.



EL Machine Strategy for Player 1

EV claim that for any pair $(v_1, v_2) \in V = (0, 1]^2$, there are strategies that achieve the pair. To see this, fix values $V_C^i, V_D^i \in (0, 1]$, with $V_C^i > V_D^i$. We now show that we can find probabilities φ^j , such that (i) in any period where j plays C , i is indifferent between C, D and obtains continuation payoff V_C^i , and (ii) in any period where j plays D , i is indifferent between C, D and obtains continuation payoff V_D^i . This requires finding four probabilities $\varphi^j \in [0, 1]$ to satisfy the four equations:

$$\begin{aligned} V_C^i &= (1 - \delta) + \delta [\varphi_{cc}^j V_C^i + (1 - \varphi_{cc}^j) V_D^i] \\ V_C^i &= (1 - \delta)2 + \delta [\varphi_{cd}^j V_C^i + (1 - \varphi_{cd}^j) V_D^i] \\ V_D^i &= (1 - \delta)(-1) + \delta [\varphi_{dc}^j V_C^i + (1 - \varphi_{dc}^j) V_D^i] \\ V_D^i &= (1 - \delta)0 + \delta [\varphi_{dd}^j V_C^i + (1 - \varphi_{dd}^j) V_D^i] \end{aligned}$$

These can be solved if δ is near 1. In particular, note that:

$$\begin{aligned} \delta(\varphi_{dc} - \varphi_{dd}) [V_C^i - V_D^i] &= (1 - \delta) \\ \delta(\varphi_{cc} - \varphi_{cd}) [V_C^i - V_D^i] &= (1 - \delta). \end{aligned}$$

As an example, let $V_C^i = 1$ and $V_D^i = 1/2$, and $\delta = 0.9$. Then we need:

$$\varphi_{cc} - \varphi_{cd} = \varphi_{dc} - \varphi_{dd} = \frac{2}{9}$$

and also from the first and last value function equations:

$$\begin{aligned} 1 &= 0.1 + 0.9 \cdot 0.5 \cdot (1 + \varphi_{cc}) \\ 0.5 &= 0.9 \cdot 0.5 \cdot (1 + \varphi_{dd}) \end{aligned}$$

which gives:

$$\begin{aligned} \varphi_{cc} &= 1 \\ \varphi_{cd} &= 7/9 \\ \varphi_{dc} &= 3/9 \\ \varphi_{dd} &= 1/9 \end{aligned}$$

Similar calculations can be done for any $1 \geq V_C > V_D \geq 0$ (provided we can pick δ near enough to 1). So any payoff in the unit square can be obtained in an equilibrium where players are indifferent between actions after every history.

Essentially the same construction continues to work if monitoring is imperfect ($\varepsilon > 0$). To prove a folk theorem for the Prisoners' Dilemma, P-EV must show that payoffs that are in the feasible strictly individually rational set, but not in the unit square can also be obtained. To do this they add a preliminary phase of finite length and support different payoffs in this preliminary phases with different continuation payoffs from the unit square. This gives the folk theorem for the case where monitoring is nearly perfect.

Matsushima (2002) uses the P-EV belief-free approach to show that if signals are conditionally independent, a Folk Theorem holds in the prisoner's dilemma even if monitoring is not nearly perfect. More recently, Ely, Horner and Olszewski (2004) characterize the entire set of payoffs that can be obtained with a belief-free equilibrium. They show that these equilibria cannot in general generate all feasible and individually rational payoffs even when players are very patient. So proving a general Folk Theorem for games with private monitoring will require something more.

3 Comments

1. Ely, Horner and Olszewski show that belief-free equilibria can be characterized in a manner similar to APS self-generation, which lends them

some appeal. A natural question, however, is whether these equilibria are fragile, in the sense that players may have to alter their mixing behavior depending on the history of play even if their continuation payoffs do not change. Bhaskar, Mailath and Morris (2004) alleviate this concern somewhat by showing that in the prisoner’s dilemma case, it is possible to purify the EV equilibria.

2. Compte (1998) and Kandori and Matsushima (1998) suggest a very different approach to private monitoring. They allow the players to communicate through cheap talk following each period. They study perfect public equilibria where the players condition on the public announcements. Both papers prove elegant extensions of the FLM folk theorem to private monitoring games with communication.
3. Horner and Olszewski (2006) prove the most general Folk Theorem to date, for general games with private almost-perfect monitoring. Their approach is to break the game into T-period blocks and construct an equilibrium where player’s beliefs matter only within each block but not across blocks. The equilibrium construction in their paper is not a trivial accomplishment.

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